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Prof. John Smith
Department of Engineering
Technical University
Berlin, 10623
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Application for Senior Research Position in Computational Engineering

Dear Prof. Smith,

I am writing to express my strong interest in the Senior Research Position in Computational Engineering at your department, as advertised on the university careers portal. With over eight years of experience in computational fluid dynamics and renewable energy systems, I believe my expertise aligns closely with the research priorities of your group. During my current appointment as a postdoctoral researcher at the Institute of Engineering Thermodynamics, I have led a team of five researchers investigating advanced simulation methods for thermal energy storage. Our work has resulted in twelve peer-reviewed publications, including three in high-impact journals, and has contributed directly to two funded research projects totalling €1.2 million. I have also supervised four master's theses and mentored two doctoral candidates through their qualifying examinations. Prior to this role, I completed my doctoral studies at TU Hamburg, where my dissertation on multi-physics modeling of solid-state thermal storage received the Faculty Award for Outstanding Research. I developed novel coupling algorithms that reduced simulation runtimes by 40% while maintaining accuracy within 2% of fully-resolved reference solutions. These methods have since been adopted by three collaborating research groups across Europe. I am particularly drawn to your department's interdisciplinary approach to engineering challenges, especially the emerging collaboration with the Materials Science department on

high-temperature energy systems. I am eager to contribute my expertise in numerical methods and thermal modeling to advance this line of research, while also expanding my own skill set through collaboration with your team.

Sincerely,

Dr. Jane Doe
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